Lesson 2: Key Press Reactor

GameMaker

VIDEO GAME CREATION

This week: sequences, events, and the game loop. Your job is to build something that reacts when you press a key - and this is also the start of your Semester 1 project, so save it somewhere safe and give it a name you'll recognize.

Tip: this lesson's Assets folder already has a ready-to-use placeholder character image (and an optional sound effect for Level 3) - you don't need to find or make your own art today.

Setup

1. Create a new blank project named [YourName]Game.
2. Create a sprite spr_player by importing one of the placeholder images from this lesson's Assets folder (e.g. player_circle_blue.png).
3. Create an object obj_player and assign spr_player to it.

Level 1 - Minimal (MVP)

One key press causes one visible reaction. This is a complete finish for today - reaching this is enough to move on to next week.

4. Add a Step event to obj_player.
5. Add the code below to the Step event.
6. Drag obj_player into your room, then press F5 (Run) to test with the Right arrow key.

```
if (keyboard_check(vk_right)) x += 4;
```

Alternate Path: GML Visual (no typing)

GameMaker lets you build the same Step event using drag-and-drop blocks instead of typed code. Pick whichever path feels right for you - both are graded the same.

7. Add a Step event to obj_player (same as the code path above).
8. In the Step event's editor, click the Code/Visual toggle in the top-right corner of the panel and switch to GML Visual.
9. From the action palette's Mouse and Keyboard category, drag an "If Key Down" block into the event and set its Key dropdown to Right.
10. From the Movement category, drag "Jump To Point" inside the If block. Check the Relative box for both X and Y, then set X to 4 and Y to 0 - Jump To Point is the dedicated block for moving relative to your current position, so prefer it over the generic Assign Variable block whenever you're just changing x/y.

11. Drag obj_player into your room, then press F5 (Run) to test with the Right arrow key - same result as the code path.

Level 2: add one more "If Key Down" + "Jump To Point" (Relative) pair per direction - X -4/Y 0 for Left, X 0/Y -4 for Up, X 0/Y 4 for Down. Level 3: reach for the most specific, well-named block GameMaker offers before falling back to Assign Variable - "Set Sprite" (Instance category) for the recolor goal instead of Assign Variable on sprite_index, and "If Audio Is Playing" -> Not -> "Play Audio" (Audio category) as the retrigger guard for the sound goal, with "Stop Audio" in the idle branch. Assign Variable is still the right tool for anything without a dedicated block, like the custom blink_timer or moving variables - same ideas as the code path either way, just dragged instead of typed.

Level 2 - Complete (Alpha/Beta)

All 4 arrow keys move your character in 4 directions.

- Add the other 3 directions using vk_left, vk_up, and vk_down.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Animate: make the sprite blink while moving, and stay solid (visible) when idle - toggle the built-in visible variable using a counter incremented in the Step event.
- Sound: play sfx_blip.wav in a beep-beep pattern while moving, silent when idle - check audio_is_playing() before calling audio_play_sound() again, and audio_stop_sound() when idle.
- Recolor: swap in a different sprite for each direction (sprite_index), rather than tinting one - import the other 4 placeholder circles as additional sprites, and set sprite_index to the matching one per direction, back to your default/orange sprite when idle.
- Clamp: keep x/y within room_width and room_height using clamp().
- Structure: keep all four of the above together in the same Step event - not scattered across separate events.

If Something's Wrong

- Nothing moves: confirm the code is in the Step event, not Draw or Create - Step runs every frame, Draw only draws.
- Confirm the obj_player instance is actually placed in the room you're running (check the Room Editor's instance list).

Want More?

gamemaker.io/en/tutorials, Student Resource List > GameMaker